automated deployment
 comparing approaches 429
 deploying customized AMIs created by Packer 422 – 429
 in-place deployment with CodeDeploy 412 – 418
 rolling updates with CloudFormation and user data 418 – 422
autoscaling
 based on CloudWatch metrics 474 – 476
 based on schedules 473 – 474
 decoupling dynamic EC2 instance pools 477 – 487
 managing dynamic EC2 instance pools 468 – 472
caching
 access control 309 – 310
 cache deployment options 302 – 308
 creating cache clusters 298 – 302
 installing sample application with CloudFormation 311 – 319
 monitoring caches 319 – 321
 tweaking performance 321 – 323
cost of 13 – 16
 billing example 14 – 15
 budget alerts 32 – 36
 Free Tier 14
 pay-per-use pricing model 16
decoupling infrastructure
 asynchronous decoupling with message queues 399 – 408
 synchronous decoupling with load balancers 393 – 399
Elastic File System
 backing up data 261 – 264
 creating filesystems 246 – 248
 creating mount targets 248 – 250
 mounting EFS filesystems on EC2 instances 250 – 254
 sharing files between EC2 instances 254 – 255
 tweaking performance 255 – 261
example applications of 5 – 10
 batch processing infrastructure 9 – 10
 highly-available system implementation 9
 running Java EE applications in private networks 7 – 8
 web shop hosting 6 – 7
fault tolerance
 considerations for 438 – 441
 Imagery 441 – 463
 redundant EC2 instances for increased availability 434 – 438